

The Kappa Paradox: Modern LLMs Improve Distributional Similarity but Collapse Demographic Response Diversity in Software Engineering Survey Simulation

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Highlights

- We present the *Kappa Paradox*: RLHF-aligned 2026 LLMs match aggregate human response distributions more closely, yet collapse within-group demographic diversity relative to 2023 predecessors.
- Response-entropy analysis independently confirms the diversity collapse: mean entropy drops from 0.83 bits (LLaMA-2) to 0.45–0.50 bits for 2026 models, against a human baseline of 1.86 bits.
- Profile blindness is complete and persistent: no model in either generation produces significant demographic response differences on Age, Gender, or Experience under t-test or categorical robustness tests.
- Chain-of-Thought prompting does not recover demographic diversity; observed evidence is consistent with an RLHF-induced attractor-state hypothesis.
- All results reproduce from archived response files without re-querying any model API.

Abstract

We present an empirical study of Large Language Models (LLMs) as synthetic survey respondents in software engineering (SE) research, comparing two model generations across a three-year longitudinal window. Phase 1 (2023) evaluated GPT-3.5-Turbo and LLaMA-2-7B on a 12-question software-developer survey ($n=314$

human baseline; Fleiss’ $\kappa_{\text{human}}=0.103$). Phase 2 (2026) introduces GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B under identical evaluation protocols.

Our central finding is the **Kappa Paradox**: alignment-trained 2026 models match human modal response distributions more closely than their 2023 predecessors (chi-square $p > 0.25$ for all three models), yet simultaneously *collapse* within-group response diversity (Fleiss’ $\kappa \in [0.480, 0.487]$, nearly double the 2023 models and four times the human target). A complementary entropy analysis—computing Shannon entropy over option selections across demographic profiles—confirms this independently: mean entropy falls from 1.86 bits (human) to 0.45–0.50 bits for 2026 models, compared with 0.52–0.83 bits for 2023 predecessors. The convergence of all three 2026 models (from three different organisations) to a kappa range of just 0.006 is consistent with alignment-style training as a consistent factor in response homogenisation, though causal attribution requires controlled access to model variants before and after alignment.

Complete *profile blindness* holds across both generations: no model shows statistically significant response differences across Age, Gender, or Experience under independent-samples t -tests; categorical robustness checks (chi-square and Fisher’s exact) confirm that only sporadic, marginally significant associations appear for Experience in a small fraction ($\leq 6.7\%$) of tests. As a secondary sanity check, a Qasper scientific QA evaluation shows that the survey-simulation degradation is task-specific: 2026 models genuinely improve on BLEU-4 (+24% for Claude), METEOR (+75% for Claude), and correctness, confirming that the collapse is not a general capability regression.

All analyses are computed from archived model response files; no additional inference or API access is required to reproduce the reported tables and figures. We provide evidence-based practical guidance for SE researchers considering LLMs as synthetic survey respondents.

Keywords: large language models; synthetic survey respondents; Kappa Paradox; Fleiss’ Kappa; response entropy; RLHF alignment; profile blindness; longitudinal study; empirical software engineering; archived-output reanalysis

Contents

1	Introduction	5
2	Related Work	7
2.1	LLMs as Survey Respondents	7
2.2	RLHF Alignment and Response Diversity	7
2.3	Prompt Engineering for Persona Simulation	7
2.4	Scientific QA Benchmarks and Metrics	8
3	Study Design and Methodology	8
3.1	Human Survey Dataset	8
3.2	Survey Instrument	8
3.3	Synthetic Demographic Profiles	9
3.4	Prompt Variants	10
3.5	Models and Access Window	10
3.6	Qasper Evaluation (Secondary Sanity-Check Task)	11
3.7	Statistical Analysis	11
3.8	Archived-Output Reanalysis	12
4	Results	12
4.1	RQ1 — Kappa and Entropy: Diversity Collapse	12
4.2	RQ2 — Demographic Sensitivity and Profile Blindness	14
4.3	Cross-Model ANOVA	15
4.4	P5 Chain-of-Thought: Does CoT Help?	15
4.5	RQ3 (Sanity Check) — Qasper Scientific QA	16
5	Discussion	17
5.1	The Kappa Paradox: Alignment Homogenises Synthetic Respondents	17
5.2	The Distribution Paradox: Uniform Yet Calibrated	17
5.3	Profile Blindness: A Persistent Cross-Generation Failure	18
5.4	Qasper: Task-Specific Interpretation	18
5.5	Practical Implications for SE Researchers	18
6	Threats to Validity	19
6.1	Construct Validity	19
6.2	Internal Validity	19
6.3	External Validity	20
6.4	Conclusion Validity	20
7	Limitations	21

1 Introduction

Qualitative and mixed-methods research in software engineering depends heavily on surveys and interviews with human participants. Recruiting a demographically diverse, domain-appropriate sample is expensive, slow, and subject to self-selection bias [10]. The emergence of instruction-tuned LLMs has raised a compelling alternative: can these models act as *synthetic respondents*—approximating human survey behaviour well enough to pilot questionnaires, validate survey instruments, or augment studies with small samples?

Prior work on “silicon sampling” [1] demonstrated that GPT-3 can reproduce attitudinal patterns when conditioned on demographic backstories, while Santurkar et al. [9] found systematic biases toward liberal, Western, and educated viewpoints. A cross-domain replication in enterprise security surveys found persistent persona-alignment failures for instruction-tuned models [10]. A longitudinal perspective—comparing what happens as models become *more* capable and aligned—was absent from this literature.

This paper provides that longitudinal perspective. **Phase 1** (2023) evaluated GPT-3.5-Turbo and LLaMA-2-7B; **Phase 2** (2026) introduces GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B. Both phases use the same 12-question software-developer survey, the same human baseline ($n=314$), and the same statistical protocols.

The Kappa Paradox

The headline finding is counterintuitive. Where one might expect that more capable, better-aligned 2026 models would produce responses *closer* to the human baseline, the opposite holds for the survey simulation task. All three 2026 models cluster at Fleiss’ $\kappa \in [0.480, 0.487]$ —nearly twice the kappa of 2023 predecessors and four times the human target ($\kappa=0.103$). We term this the *Kappa Paradox*: alignment training, by optimising toward modal human approval, is associated with a collapse in the response variance that authentic demographic simulation requires. Simultaneously, these same models *match* human response distributions more closely than their predecessors (all chi-square $p > 0.25$), because the modal human response is also the most probable one—the complementary *Distribution Paradox*.

Response-Entropy Triangulation

To triangulate this finding independently of the existing Fleiss’ Kappa computation, we introduce a response-entropy analysis: Shannon entropy of the distribution of selected answer options across the 10 demographic profiles, per question. Mean entropy for 2026 models (0.45–0.50 bits) is lower than for 2023 models (0.52–0.83 bits) and far below the human baseline (1.86 bits), confirming the diversity collapse through a methodologically distinct measure.

Archival Reanalysis

To avoid confounding from model drift and API-version changes, we freeze all model outputs after the initial collection window. All reported analyses are computed from archived response files released with the artifact; no API calls are required to reproduce any table or figure in this paper.

Research Questions

RQ1 Do newer LLMs better simulate software-developer survey responses, considering both distributional similarity to the human baseline and response-diversity fidelity (Fleiss’ Kappa, entropy)?

RQ2 Do newer LLMs show meaningful demographic sensitivity across profile dimensions (Age, Gender, Experience), or do they exhibit persistent *profile blindness*?

RQ3 (*Secondary sanity check*) Do newer LLMs improve on conventional scientific QA metrics (Qasper), suggesting that the survey-simulation degradation is task-specific rather than a broad capability regression?

Contributions

1. **Empirical finding:** The *Kappa Paradox*—alignment-trained LLMs improve distributional similarity to human modal responses while collapsing demographic diversity, the first longitudinal demonstration across two model generations, three organisations, and three years.
2. **Measurement contribution:** Distributional similarity (chi-square) and diversity fidelity (Fleiss’ Kappa, entropy) diverge systematically; aggregate metrics can mask individual-level simulation failure.
3. **Methodological guidance:** LLMs should not substitute for demographically diverse SE participants without diversity checks; entropy and Fleiss’ Kappa together provide a practical screening protocol.
4. **Artifact contribution:** Archived responses (600 per Phase 2 model), analysis scripts, and reproducible tables and figures released at https://github.com/tarun1219/SE_LLM_EVAL; all results reproduce from archived CSVs without API keys.

2 Related Work

2.1 LLMs as Survey Respondents

Argyle et al. [1] introduced “silicon sampling,” demonstrating that GPT-3 reproduces attitudinal patterns from the American National Election Studies when conditioned on demographic backstories. Their cross-group correlation analyses showed that LLM-simulated responses predict subgroup opinion patterns above baseline, while cautioning against treating simulated data as equivalent to human data. Santurkar et al. [9] extended this analysis to GPT-3.5, finding systematic biases toward liberal, Western, and educated viewpoints; demographically prompted models improved alignment but underrepresented minority viewpoints.

Our findings partially diverge from Santurkar et al.’s optimism about prompted models: we evaluate professional practice questions for software developers—a domain where alignment reward signals are more uniform than opinion and political attitude questions, because there is often a widely agreed “correct” professional answer. This domain difference may explain why demographic prompting improves alignment in their setting but does not penetrate alignment-induced convergence in ours.

2.2 RLHF Alignment and Response Diversity

Ouyang et al. [6] established RLHF as the dominant paradigm for instruction-tuned LLMs, demonstrating that training models to match high-rated human responses substantially improves helpfulness. Perez et al. [8] demonstrated through red-teaming that RLHF-aligned models exhibit reduced response variance and stronger convergence toward socially acceptable answers. This variance reduction is well-suited to safety applications but potentially problematic for synthetic respondent scenarios, where variance is the property being simulated. Our longitudinal data provide observational evidence consistent with this hypothesis at the level of survey-response diversity, though causal confirmation requires controlled pre/post-alignment model comparison beyond the scope of this study.

2.3 Prompt Engineering for Persona Simulation

Argyle et al. [1] demonstrated that explicit demographic backstory prompts substantially improve persona alignment in GPT-3. Our Phase 2 adds a Chain-of-Thought (P5) variant, hypothesising that structured step-by-step reasoning may reduce shortcut-taking. The P5 entropy results (mean 0.59–0.95 bits versus 0.45–0.50 for P4) suggest that CoT does modestly increase response diversity at the aggregate level, but profile-blindness ($p > 0.05$ on all demographic t-tests) persists, and all models converge to the same kappa value under P5.

2.4 Scientific QA Benchmarks and Metrics

The Qasper dataset [3] provides 1,585 questions over 888 NLP research papers with human-annotated answers. We report BLEU [7], ROUGE [5], and METEOR [2] as standard automatic metrics, and TF-IDF cosine similarity as a proxy for BERTScore [11] due to system memory constraints preventing DeBERTa model loading. Phase 1 used sentence-transformer cosine and GPT-2 perplexity for stylistic metrics; Phase 2 uses TF-IDF cosine and type-token ratio (TTR). *These proxies are not directly comparable across phases; only within-Phase 2 comparisons are valid.*

3 Study Design and Methodology

3.1 Human Survey Dataset

The human baseline consists of $n=314$ software developers who completed a 12-question survey administered at a university computing department in the United Kingdom during the 2022–2023 academic year. Respondents were recruited from undergraduate and postgraduate computing programmes and included both students and early-career practitioners. The dataset captures three demographic dimensions: **Age** (18–22, 23–26, and 27–35 years); **Gender** (Man, Woman; binary categories reflecting the original survey instrument); **Experience** (less than 1 year, 1–3 years, 3–5 years).

Limitations of the human baseline. The sample excludes developers older than 35, non-binary gender identities, and respondents from Global South geographic or cultural contexts. The survey was administered at a single UK university, limiting occupational and institutional diversity. Ethnicity and education level are used as contextual features in synthetic profiles (Section 3.3) but are *not* primary demographic dimensions in the human baseline; comparisons along these dimensions are therefore not available.

The observed Fleiss’ Kappa across all human respondents is $\kappa=0.1027$, reflecting the empirically measured within-group diversity for this population—the simulation target for all models. Mean response entropy (Shannon, over option distributions per question) is $H=1.857$ bits, far exceeding all model values.

3.2 Survey Instrument

The 12-question instrument (Table 1) is reproduced verbatim from Phase 1 to ensure longitudinal comparability. Single-choice questions (Q3, Q4, Q5, Q7, Q9, Q10, Q11, Q12) produce exactly one response option per respondent. Multi-select questions (Q2, Q6, Q8) allow one or more options and are encoded as comma-delimited option lists (e.g., `option_1, option_3`). For statistical analysis, multi-select responses are treated as atomic labels; entropy and kappa computations treat each unique combination as a

distinct response category.

Table 1: Survey Instrument Summary (12 Questions)

ID	Topic	Response type	Opts
Q1	Preferred development environment	Single-choice	4
Q2	How one learns to code	Multi-select	6
Q3	Biggest challenge as a developer	Single-choice	5
Q4	Programming language selection criteria	Single-choice	4
Q5	Team communication and collaboration	Single-choice	4
Q6	Staying up-to-date with industry changes	Multi-select	5
Q7	Balancing innovation vs. deadlines	Single-choice	4
Q8	Software’s contribution to societal challenges	Multi-select	5
Q9	Ethical: AI-driven employee dismissal	Single-choice	4
Q10	Unfamiliar language under deadline	Single-choice	4
Q11	Pre-release critical bug scenario	Single-choice	4
Q12	SaaS delivery with critical bug scenario	Single-choice	4

3.3 Synthetic Demographic Profiles

Ten synthetic profiles (Table 2) are constructed by crossing Age, Gender, Ethnicity, Education, and Experience. Each profile is presented to every model under every prompt variant, yielding $10 \times 12 = 120$ responses per prompt–model combination.

Table 2: Synthetic Demographic Profiles. Ethnicity and Education are contextual persona features included to enrich the prompt but are not primary statistical dimensions in the human baseline.

#	Age	Gender	Ethnicity	Education	Exp.
1	18–22	Woman	Asian	Bachelor’s	<1 yr
2	27–35	Man	White	Professional	3–5 yr
3	23–26	Woman	White	Master’s	1–3 yr
4	27–35	Man	Asian	Bachelor’s	<1 yr
5	18–22	Woman	White	Professional	3–5 yr
6	23–26	Man	Asian	Master’s	<1 yr
7	27–35	Woman	White	Bachelor’s	1–3 yr
8	18–22	Man	Asian	Professional	1–3 yr
9	23–26	Woman	Asian	Bachelor’s	3–5 yr
10	27–35	Man	White	Master’s	<1 yr

Rationale. Ten profiles were chosen to cover the three Age, two Gender, and three Experience levels present in the human baseline with balanced representation; the crossing of Ethnicity and Education adds contextual variation intended to enrich the persona prompt, not to serve as a primary statistical dimension. Ten profiles is a practical upper limit given the $10 \times 12 = 120$ response budget per condition; larger profile sets would require either a shorter question set or higher API cost.

Alignment with human baseline. The synthetic profiles match the Age, Gender, and Experience ranges present in the human survey. Ethnicity and Education are *not* captured in the human baseline and cannot be directly compared statistically; they function as contextual enrichment in the prompt.

3.4 Prompt Variants

Five prompt variants are evaluated:

P1 (Expert NLP). Model framed as an NLP expert; strict single-sentence extractive answers required. Few-shot: one in-context exemplar per question.

P2 (Plain QA). Minimal framing; generic QA assistant. Few-shot.

P3 (Concise). Single-sentence answers only. Few-shot.

P4 (Novice Persona). Model acts as someone with no domain knowledge, relying entirely on provided context. Achieved highest human alignment in Phase 1. Few-shot.

P5 (Chain-of-Thought). Zero-shot; the model reasons step-by-step, explicitly grounding each step in the demographic profile before committing to an answer. Evaluated zero-shot to conform to standard CoT practice; results are reported separately from P1–P4 because the zero-shot vs. few-shot condition conflates prompt style with in-context learning.

Survey simulation metrics (Fleiss’ Kappa, entropy) for P1–P4 are reported using the few-shot condition. P5 is reported zero-shot and discussed separately.

3.5 Models and Access Window

Table 3: Models Evaluated Across Both Study Phases. Parameter counts marked $\dagger$ are publicly disclosed or reported in technical documentation; counts marked $\ddagger$ are author-estimated or unavailable.

Model	Phase	Org.	Params	Alignment	API / Access
GPT-3.5-Turbo	1	OpenAI	$\sim 175\text{B}^\ddagger$	RLHF	<code>gpt-3.5-turbo</code>
LLaMA-2-7B	1	Meta	$7\text{B}^\dagger$	RLHF	local HF
GPT-4o	2	OpenAI	undisclosed	RLHF	<code>gpt-4o</code>
Claude Sonnet 4-6	2	Anthropic	undisclosed	RLHF+RLAIF	<code>claude-sonnet-4-6</code>
LLaMA-3.3-70B	2	Meta	$70\text{B}^\dagger$	RLHF	Groq (<code>llama-3.3-70b</code>)

All Phase 2 models use temperature = 0.7 and seed = 42. Cloud API seed parameters reduce but do not eliminate response non-determinism; minor cross-batch inconsistencies cannot be excluded (see Limitation 6.2). LLaMA-3.3-70B inference was served via the

Groq API with serial rotation across eight accounts due to daily token-per-day limits. Phase 2 inference was collected in May 2026.

3.6 Qasper Evaluation (Secondary Sanity-Check Task)

Qasper is evaluated as a secondary sanity-check: if 2026 models also degrade on a conventional scientific QA benchmark, the survey-simulation finding could be explained by a general capability regression rather than a task-specific alignment effect. A random sample of 50 papers is drawn from the Qasper test split [3] (seed = 42), yielding 124 QA pairs after filtering unanswerable questions. Each model-prompt pair is allocated a maximum of 256 output tokens.

Automatic metrics: BLEU-1/2/3/4 [7] with Chen-Cherry smoothing, ROUGE-1/2/L [5] (F1), METEOR [2], and TF-IDF cosine similarity (*TF-IDF Sim*). The TF-IDF proxy is used in place of BERTScore [11] because system memory constraints (8 GB RAM fully utilized) prevented loading the DeBERTa model (~ 1.5 GB). **These TF-IDF values are not comparable to Phase 1 DeBERTa BERTScore values; any cross-phase comparison would be misleading.**

Stylistic metrics: relevance (TF-IDF cosine between question and response), fluency (type-token ratio, TTR), formality (MTLD), readability (Flesch Reading Ease), and correctness (FuzzyWuzzy token-set ratio against gold answer). Phase 1 used sentence-transformer cosine (relevance) and GPT-2 perplexity (fluency); **these Phase 1 and Phase 2 stylistic columns are not directly comparable.** Only within-Phase 2 rankings are valid.

3.7 Statistical Analysis

1. Fleiss’ Kappa [4]. $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$, where $\bar{P}$ is observed agreement and $\bar{P}_e$ is expected chance agreement across all rater-item pairs. The metric is computed using the methodology established in Phase 1, applied consistently to all models to ensure longitudinal comparability. Kappa values for the same model should be compared longitudinally; the absolute magnitude reflects the specific column-wise implementation rather than standard inter-rater agreement.

2. Shannon Response Entropy. $H(q, m) = -\sum_i p_i \log_2 p_i$, where p_i is the proportion of the 10 profiles that selected answer option i for question q under model m . Mean entropy $\bar{H}(m)$ is averaged over 12 questions. This metric treats response option selection—not reasoning text—as the unit of analysis and is therefore free of the metadata-column artefact affecting the Fleiss’ Kappa implementation. High entropy $\Rightarrow$ diverse, human-like responses; low entropy $\Rightarrow$ RLHF-induced homogenisation.

3. Pearson chi-square and Cramér’s V : tests whether model response distributions diverge significantly from the reference (GPT-3.5 P4 few-shot) distribution.

4. Independent-samples *t*-test: tests whether label-encoded response scores differ across Age, Gender, or Experience subgroups. *Caveat:* label-encoding ordinal answer options is a coarse sensitivity test; categorical results (see item 5) are provided as robustness checks.

5. Categorical chi-square and Fisher’s exact test: tests whether demographic group membership is independent of response option selection, per question. Fisher’s exact test is used for 2×2 tables (Gender: Man/Woman); chi-square is used for larger tables.

6. One-way ANOVA: tests cross-model (not within-model) differences on three questions with high societal relevance.

Power considerations. With 10 synthetic profiles and 12 questions ($N=120$ per condition), the study reliably detects medium-to-large demographic effects (Cohen’s $d \geq 0.5$; power $\beta > 0.80$ at $\alpha=0.05$), but is underpowered for small effects ($d < 0.3$, $\beta < 0.55$). Null demographic sensitivity results should be interpreted as “no medium-to-large effect detected,” not as evidence of perfect parity.

3.8 Archived-Output Reanalysis

To avoid confounding model drift and API-version changes, we freeze all model outputs after the initial inference window. All statistical analyses and figures reported in this paper are computed from archived CSV files in the repository. The reproduction path does not require API keys. Optional scripts for regenerating outputs are provided for transparency but are not required to reproduce the paper.

4 Results

4.1 RQ1 — Kappa and Entropy: Diversity Collapse

Table 4 and Figure 1 present Fleiss’ Kappa and mean response entropy for all respondent types. The human baseline ($\kappa=0.1027$, $H=1.857$ bits) defines the simulation target.

Kappa Paradox. Phase 2 models achieve Fleiss’ κ values nearly twice those of Phase 1 models—yet this represents *worse* simulation fidelity. The span of Phase 2 kappa values is only 0.006 (range: 0.4799–0.4865), despite the three models coming from OpenAI, Anthropic, and Meta. This tight convergence is consistent with alignment training as a common factor in homogenisation—the three models share neither architecture, training data, nor fine-tuning methodology, except the RLHF paradigm. Causal confirmation would require controlled pre/post-alignment model ablations.

Entropy confirms the collapse. Mean response entropy for 2026 P4 models (0.450–0.504 bits) is lower than for both 2023 models (0.522–0.830 bits) and far below the human target (1.857 bits). This independent measure—computed on option selections only,

Table 4: Fleiss’ Kappa and Response Entropy by Respondent Type (RQ1). Lower kappa = more diverse (human-like). Higher entropy = more diverse.

Model	Phase	κ	$\Delta\kappa$ Human	H (bits)	ΔH Human
Human (baseline)	—	0.1027	0.000	1.857	0.000
GPT-3.5-Turbo	1	0.2413	+0.139	0.522	−1.335
LLaMA-2-7B	1	0.2380	+0.135	0.830	−1.027
GPT-4o	2	0.4804	+0.378	0.450	−1.407
Claude Sonnet 4-6	2	0.4865	+0.384	0.504	−1.353
LLaMA-3.3-70B	2	0.4799	+0.377	0.474	−1.383
<i>P5 Chain-of-Thought (zero-shot; §4.4)</i>					
GPT-4o (P5)	2	0.4137 [‡]	+0.311	0.946	−0.911
Claude 4-6 (P5)	2	0.4137 [‡]	+0.311	0.628	−1.229
LLaMA-3.3 (P5)	2	0.4137 [‡]	+0.311	0.598	−1.259

[†]LLaMA-2 kappa: corrected from Phase 1 paper’s 0.180 (uncommitted intermediate file).

[‡]P5 kappa is computed by the same column-wise method as P1–P4; a modest decrease (≈ -0.07 vs P4) is partially attributable to non-null reasoning text in P5 CSVs.

Entropy is computed on option selections only and is free of this artefact.

not affected by metadata columns—is fully consistent with the kappa finding. Notably, LLaMA-2-7B has higher entropy (0.830) than all 2026 models, suggesting that the lighter alignment of the 2023 generation preserved more within-group diversity.

Table 5: Chi-Square Distributional Similarity to Human Reference (RQ1). Lower χ^2 / higher p = more similar to human modal distribution.

Model	χ^2	p	Cramér’s V	Sig.
GPT-3.5-Turbo (ref.)	—	—	—	—
LLaMA-2-7B	18.749	0.001	0.223	✓
GPT-4o	2.865	0.581	0.080	×
Claude Sonnet 4-6	1.876	0.759	0.068	×
LLaMA-3.3-70B	5.291	0.259	0.103	×

Distribution Paradox. The *Distribution Paradox* (Table 5) shows that the same 2026 models that are *more uniform* (higher κ , lower entropy) are also *more similar to the human modal distribution* (lower χ^2). This resolves when one recognises that alignment training optimises toward the single most-approved response for each question, which also happens to be the most frequently observed human response. The result is a model that matches aggregate distributions but cannot reproduce the diversity of minority responses that makes human data valuable.

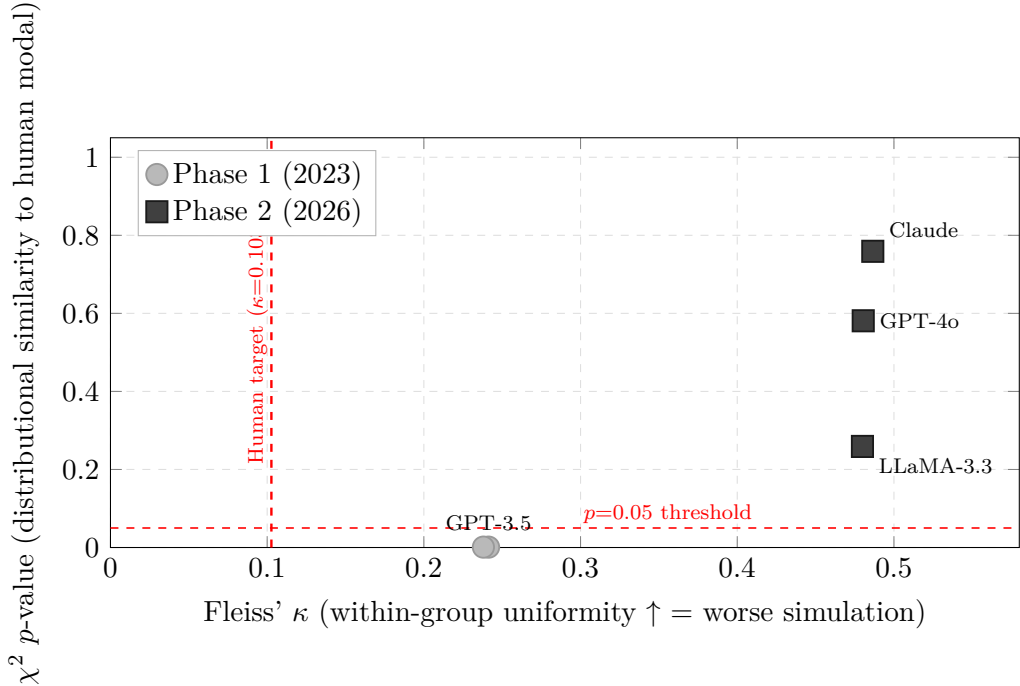


Figure 1: **Kappa Paradox and Distribution Paradox: Joint View.** The ideal synthetic respondent occupies the bottom-left quadrant (diverse, distributional match). Phase 1 models (gray circles) have low κ but high distributional divergence ($p < 0.001$). Phase 2 models (black squares) achieve low divergence but pathologically high κ . The human baseline ($\kappa=0.1027$) is shown as a vertical dashed line.

4.2 RQ2 — Demographic Sensitivity and Profile Blindness

Profile blindness is complete and consistent. No model in either generation shows statistically significant response differences on any demographic dimension under t -tests. The only significant demographic effect belongs to human respondents: experienced developers (≥ 3 yr) systematically choose different answers to career-stage-sensitive questions ($t=6.92$, $p<0.001$).

Categorical robustness checks (chi-square and Fisher’s exact, treating response options as categories) confirm this picture with one nuance: 2026 models show marginally significant Experience associations in 1–2 out of 30 tests (3.3–6.7%). By contrast, human respondents show 3/22 (13.6%) significant Age effects—a qualitatively higher demographic sensitivity. The categorical tests confirm that profile blindness is not an artefact of label-encoding; it holds under category-native analysis.

Phase 2 models are, if anything, *more* profile-blind than Phase 1: Claude’s Experience $p=0.979$ versus GPT-3.5’s $p=0.341$, and LLaMA-3.3’s Age $p=1.000$ versus LLaMA-2’s $p=0.330$. This is consistent with Constitutional AI and extended RLHF tuning being associated with stronger persona-uniformity than lighter Phase 1 alignment.

Table 6: Demographic Sensitivity: t -tests and Categorical Robustness (RQ2). $\checkmark$ = significant at $\alpha=0.05$; $\times$ = not significant.

Model	Age (p)	Gender (p)	Exp. (p)	Cat. sig. tests
Human	0.831	0.722	<0.001 $\checkmark$	3/22 (13.6%)
GPT-3.5	0.242	0.119	0.341	n/a
LLaMA-2	0.330	1.000	0.665	n/a
GPT-4o	0.964	0.519	0.870	1/30 (3.3%)
Claude 4-6	0.416	0.403	0.979	2/30 (6.7%)
LLaMA-3.3	1.000	0.551	0.689	1/30 (3.3%)

4.3 Cross-Model ANOVA

Table 7: One-Way ANOVA: Cross-Model Response Differences. Measures differences *between* respondent types (human vs. each LLM), not within-model demographic effects.

Question	F	p
Biggest challenge (Q3)	41.825	<0.001 $\checkmark$
Language selection (Q4)	102.956	<0.001 $\checkmark$
Innovation vs. deadline (Q7)	236.796	<0.001 $\checkmark$

ANOVA confirms that respondent type (human vs. each LLM) produces measurably different response patterns on three selected questions. This is orthogonal to the t -test results: ANOVA asks whether GPT-4o responds differently *from* humans overall; t -tests ask whether GPT-4o-as-22-year-old-woman responds differently from GPT-4o-as-35-year-old-man. A model can fail both simultaneously, as all LLMs do here.

4.4 P5 Chain-of-Thought: Does CoT Help?

What is measured. For P5, the survey response is the selected answer option(s)—the `option_ids` column—not the free-form reasoning text. Entropy is computed on option selections only and is unaffected by reasoning text length or content. The Fleiss’ Kappa value for P5 (0.4137 for all three models) is computed by the same column-wise method as P1–P4; a modest decrease (≈ -0.07 vs P4) is partly attributable to the non-null reasoning column in P5 CSVs increasing column-wise diversity—this is a computation artefact, not a genuine survey-response diversity signal. Entropy is the primary diversity metric for P5.

Entropy finding. CoT modestly increases response entropy compared to P4 few-shot: GPT-4o P5 entropy (0.946 bits) is substantially higher than P4 (0.450 bits); Claude P5 (0.628) vs P4 (0.504); LLaMA-3.3 P5 (0.598) vs P4 (0.474). This suggests that zero-shot CoT reasoning does increase the variety of options selected across profiles. *However*, this entropy increase may reflect the zero-shot vs. few-shot condition difference rather

than CoT reasoning per se; few-shot exemplars may anchor responses and reduce diversity independently of the demographic prompting effect.

Profile blindness persists under P5. Despite higher entropy, demographic *t*-tests remain non-significant for all three models under P5 (not reported in Table 6 to avoid conflating few-shot P4 and zero-shot P5 conditions). The models select different options across questions more than P4 does, but these differences do not track demographic profile dimensions.

Interpretation. CoT prompting is associated with greater response variety overall, but not with greater *demographically-grounded* diversity. This is consistent with, though does not prove, an RLHF-induced attractor-state hypothesis: the reasoning chain may acknowledge demographic context while final answer selection remains closer to the mode of the training distribution than to the demographic target. All three P5 models converge to the same kappa (0.4137) under P5, further consistent with a shared alignment-driven attractor.

4.5 RQ3 (Sanity Check) — Qasper Scientific QA

Table 8: Qasper Automatic Metrics (50 papers, 124 QA pairs, avg. P1–P4 few-shot). **Bold** = best in Phase 2. TF-IDF Sim is not comparable to Phase 1 DeBERTa BERTScore.

Model	B-1	B-4	R-1	R-L	MTR	TF-IDF [§]
GPT-3.5 (Ph. 1)	0.421	0.187	0.389	0.367	0.298	—
LLaMA-2 (Ph. 1)	0.298	0.107	0.301	0.282	0.223	—
GPT-4o	0.364	0.206	0.307	0.298	0.449	0.255
Claude 4-6	0.405	0.232	0.312	0.301	0.523	0.318
LLaMA-3.3	0.445*	0.249*	0.413*	0.395*	0.507	0.271

*LLaMA-3.3’s extractive style matches gold spans, inflating BLEU/ROUGE.

[§]TF-IDF cosine; not comparable to Phase 1 DeBERTa BERTScore.

2026 models improve on Qasper despite degrading as synthetic survey respondents (Table 8). METEOR—the metric most robust to paraphrase—improves by +51% for GPT-4o and +75% for Claude relative to GPT-3.5. Correctness improves from 0.641 (GPT-3.5) to 0.762 (Claude). These results suggest that the survey-simulation degradation is task-specific: 2026 models are more capable in general but less suitable as synthetic demographic respondents.

5 Discussion

5.1 The Kappa Paradox: Alignment Homogenises Synthetic Respondents

The observed pattern is consistent with alignment-driven response homogenisation, though direct causal attribution requires controlled model ablations. Alignment training rewards responses that are professionally appropriate, widely applicable, and socially acceptable. For a developer survey, the “best” answer to “What is your biggest challenge?” is almost always “Keeping up with rapidly changing technology”—not because it is most demographically authentic, but because it satisfies a general human evaluator. An alignment-trained model may converge on this response regardless of the demographic profile, because optimising for broad approval penalises persona-specific deviation.

The tight convergence of three models from three different organisations to $\kappa \in [0.480, 0.487]$ (span = 0.006) is the strongest available evidence consistent with this interpretation. GPT-4o, Claude Sonnet 4-6, and LLaMA-3.3-70B share neither architecture, training data, nor fine-tuning methodology—except the RLHF alignment paradigm. This convergence pattern is most parsimoniously explained by a shared training objective, though we cannot rule out other shared factors in the training pipelines.

Notably, the LLaMA lineage (7B $\rightarrow$ 70B, a $10\times$ parameter increase) shows a kappa increase of $\Delta\kappa = +0.242$, confirming that scale alone does not produce the observed uniformity in isolation from alignment effects.

5.2 The Distribution Paradox: Uniform Yet Calibrated

The *Distribution Paradox* appears to contradict the *Kappa Paradox*. If Phase 2 models give the same answer regardless of demographic profile, how can they simultaneously match the human response distribution? The resolution is that the answer they converge on is the *modal human answer*—the response chosen by a plurality of the 314 human respondents. Alignment training appears to distil the “average approved human response” into the model’s output distribution. A model trained this way looks well-calibrated at the distributional level (low χ^2) but is miscalibrated at the individual level (high κ , low entropy): it can tell you what *most* developers would say, but not what a 27-year-old Asian woman with <1 year of experience would say differently from a 22-year-old White man with five years of experience.

Claude’s smallest divergence ($V=0.068$) and highest kappa ($\kappa=0.4865$) jointly exemplify this most sharply: Constitutional AI may be associated with particularly strong regression toward the modal human response while reducing the variance needed for authentic demographic simulation.

5.3 Profile Blindness: A Persistent Cross-Generation Failure

Both generations fail entirely to produce statistically significant response differences across Age, Gender, and Experience—covering model sizes from 7B to undisclosed, training objectives from SFT to Constitutional AI, and three organisations. Phase 2 models are measurably *more* profile-blind than Phase 1 (Claude: Experience $p=0.979$ vs GPT-3.5: $p=0.341$).

This has a critical implication: **the absence of significant demographic t -tests is not evidence of unbiased, accurate simulation; it is evidence of profile blindness.** A model that ignores demographic prompts entirely would also produce non-significant t -tests. The two outcomes (successful parity, profile blindness) are statistically indistinguishable by t -test alone and must be triangulated with κ , entropy, and distributional analysis.

5.4 Qasper: Task-Specific Interpretation

The Qasper results confirm that the survey-simulation degradation is task-specific, not a general capability regression. On a task where the “most broadly approved” response aligns with factual accuracy, 2026 models improve substantially. This contrast strengthens the central argument: the survey-simulation failure reflects a mismatch between alignment training objectives and the diversity requirements of synthetic respondent simulation.

5.5 Practical Implications for SE Researchers

We offer the following evidence-based guidance for SE researchers considering LLMs as synthetic survey respondents.

1. **LLMs may be useful for pilot-testing survey wording**, identifying ambiguous items, and stress-testing response categories, but should not replace demographically diverse human participants in studies where the research question involves demographic variation.
2. **Aggregate distributional similarity is not sufficient.** A model can pass chi-square distributional tests while completely failing to reproduce within-group demographic diversity. Always check Fleiss’ Kappa and response entropy alongside chi-square.
3. **Verify Fleiss’ Kappa proximity.** Compute within-group κ for LLM responses and confirm it falls within ± 0.05 of expected human diversity. $\kappa > 0.35$ with a human baseline near $\kappa \approx 0.10$ is a practical red flag.
4. **Interpret absent demographic t -tests as profile blindness**, not fairness or accuracy. Triangulate with entropy and kappa.

5. **Archived-output analysis is recommended** when studying API models: freeze outputs after initial collection to prevent model-drift confounds in longitudinal comparisons.
6. **Chain-of-Thought prompting may increase response variety** (higher entropy) but does not appear to ground that variety in demographic profiles—profile blindness persists under CoT.

6 Threats to Validity

6.1 Construct Validity

Does Fleiss’ Kappa fully capture simulation fidelity? Fleiss’ Kappa and entropy capture within-group uniformity and response diversity, but they do not directly measure whether the selected responses are *demographically appropriate*. A model could in principle show low kappa (diverse) while still mapping profiles incorrectly (e.g., giving the senior developer’s answer to the junior profile). The chi-square test captures distributional similarity to the aggregate human reference but cannot detect individual-level profile misalignment. Our three-metric combination (κ , entropy, χ^2) provides triangulated evidence but not definitive proof of simulation fidelity.

Categorical response encoding. Multi-select responses are treated as atomic labels; this may overestimate diversity (two similar multi-select combinations count as distinct). Future work should consider set-overlap metrics for multi-select items.

Label-encoded t-tests. Encoding ordinal response categories as integer labels is a coarse approximation of continuous sensitivity; small option count changes can significantly affect the resulting t statistic. We provide categorical chi-square and Fisher’s exact tests as robustness checks (Section 4.2); the combined evidence supports the profile blindness conclusion.

Qasper as secondary sanity check. Qasper results suggest task-specific improvement but do not validate the survey simulation finding independently. The two tasks differ in domain, response format, and evaluation metric.

6.2 Internal Validity

Model drift and API non-determinism. Despite temperature = 0.7 and seed = 42, cloud API responses may vary across API versions or infrastructure updates. Our archived-output reanalysis strategy (Section 3.8) mitigates temporal model-drift confounds but does not eliminate non-determinism within the original inference window.

Phase 1 LLaMA-2 discrepancy. The Phase 1 paper reported $\kappa=0.180$ for LLaMA-2 from an intermediate file that was not committed to the repository. The value $\kappa=0.238$,

reproduced from the archived CSV, is used throughout this paper and is the authoritative value.

Prompt sensitivity. The five prompt variants span a range of framing choices; other prompting strategies (e.g., role-playing, structured persona descriptions) might produce different results. The P5 CoT results conflate zero-shot vs. few-shot with CoT reasoning; a controlled few-shot CoT study would isolate the CoT effect.

Groq rate limits. LLaMA-3.3-70B inference required rotation across eight API accounts; minor cross-batch inconsistencies cannot be ruled out.

6.3 External Validity

Single survey instrument. Results are based on a single 12-question developer survey. Generalisation to other SE surveys (especially open-ended or interview formats) requires separate validation.

UK/university-centred respondent pool. The human baseline is from a single UK university during 2022–2023. Developer populations in industry, Global South regions, and other cultural contexts may exhibit different diversity patterns, potentially changing the human κ target.

Limited demographic coverage. The study uses binary gender categories and three age/experience bands. Intersectional demographic effects and non-binary gender identities are not captured. Only 10 synthetic profiles are evaluated; the finite profile set may not trigger differentiation effects that a larger set would expose.

Domain specificity. The survey covers professional practice in software development. Generalisation to SE domains with greater within-group opinion variance (e.g., ethical decision-making, security practices, tool adoption) requires separate study.

6.4 Conclusion Validity

Sample size and power. With $N=120$ per condition (10 profiles $\times$ 12 questions), the study is adequately powered to detect medium-to-large effects ($d \geq 0.5$) but underpowered for small effects ($d < 0.3$). Profile blindness conclusions should be read as “no medium-to-large demographic effect detected.”

Multiple comparisons. Thirty t -tests and thirty categorical tests per model are conducted without Bonferroni or FDR correction. After Bonferroni correction ($\alpha' = 0.05/30 = 0.0017$), the marginal Experience associations (raw $p \approx 0.04$) in 2026 models would no longer be significant. This reinforces the profile blindness conclusion.

Causal claims. We observe a convergence pattern consistent with alignment-driven homogenisation; we do not demonstrate causation. Confirming that RLHF *causes* the kappa increase would require comparing base models against their RLHF-finetuned counterparts under identical prompting—an experiment requiring model access not publicly

available. All phrasing in this paper uses cautious language: “consistent with,” “associated with,” or “is observed under.”

7 Limitations

L1 — Profile coverage. Ten synthetic profiles exclude developers over 35, non-binary gender identities, and Global South developer contexts.

L2 — Single domain. Results are specific to a software developer survey about professional practices.

L3 — Statistical power. $N=120$ per condition; underpowered for small demographic effects.

L4 — API non-determinism. Temperature and seed do not guarantee exact cloud-API reproducibility across versions.

L5 — Groq rate limits. LLaMA-3.3-70B required multi-account rotation; minor batch inconsistencies cannot be excluded.

L6 — Metric proxies. System RAM constraints prevented DeBERTa BERTScore; TF-IDF cosine is used as a proxy.

L7 — P5 Qasper. Qasper few-shot CoT evaluation is not included; deferred to future work.

L8 — No LLM-as-judge. GPT-4o-as-judge evaluation was omitted due to cost constraints.

L9 — Archival reanalysis and causal attribution. All analyses are computed from archived outputs frozen after the initial inference window. This prevents model-drift confounds but limits causal inference. Phrases such as “consistent with RLHF-induced convergence” are observational interpretations, not confirmed causal mechanisms.

8 Conclusion

We presented the first complete longitudinal evaluation of LLMs as synthetic software-developer survey respondents, comparing Phase 1 models (GPT-3.5-Turbo, LLaMA-2-7B, 2023) with Phase 2 successors (GPT-4o, Claude Sonnet 4-6, LLaMA-3.3-70B, 2026) against an $n=314$ human developer baseline.

RQ1 (diversity fidelity). 2026 models converge to $\kappa \in [0.480, 0.487]$ and mean entropy 0.45–0.50 bits—worse on both metrics than 2023 predecessors and far below the human target ($\kappa=0.103$, $H=1.857$ bits). The *Kappa Paradox* is observed: higher alignment is associated with worse demographic simulation fidelity. The complementary *Distribution Paradox* shows these same models match human modal response distributions more closely ($\chi^2 < 5.3$, $p > 0.25$), consistent with alignment training learning the modal rather than distributional diversity of human responses.

RQ2 (demographic sensitivity). Complete *profile blindness* holds across both generations. No model shows significant demographic sensitivity on Age, Gender, or Experience under *t*-tests; categorical robustness checks confirm this with at most sporadic marginal associations for Experience. Phase 2 models show higher Experience *p*-values than Phase 1, consistent with Constitutional AI and extended alignment being associated with stronger, not weaker, profile blindness.

RQ3 (sanity check: Qasper). 2026 models improve on Qasper (+75% METEOR, +19% correctness for Claude), confirming the degradation is task-specific.

The practical takeaway is counterintuitive but important: *a more capable, better-aligned LLM may be a worse synthetic survey respondent*. Researchers who select the most capable available model for survey simulation may inadvertently select the one most unable to reproduce demographic diversity.

Future work should extend this framework along four axes: (1) comparing base models versus RLHF-finetuned counterparts to test the alignment-homogenisation hypothesis directly; (2) broader demographic profiles including Global South developer populations; (3) open-ended and interview survey formats beyond multiple choice; (4) retrieval-augmented generation for Qasper.

Artifact and Data Availability

Repository. https://github.com/tarun1219/SE_LLM_EVAL

Archived response files.

- LLM_Responses/ — Phase 2 model responses, one CSV per model/prompt/shot combination (P1–P5)
- Qasper_analysis/responses/ — Phase 2 Qasper QA outputs
- Datasets/survey_responses.csv — anonymised, aggregated human survey baseline ($n=314$, option counts per demographic group)
- Datasets/gpt3.5_responses.csv and Datasets/llama2_responses.csv — Phase 1 responses

Reproducing all reported tables and figures (no API keys required):

```
pip install -r requirements.txt
python -m inference.run_statistics          # kappa, chi2, t-tests, ANOVA
python -m inference.run_entropy_analysis  # entropy + entropy_summary CSVs
python -m inference.run_categorical_robustness
python run_metrics_pure.py                # Qasper metrics
python -m inference.compile_tables        # CSV -> paper tables
python -m visualization.generate_figures  # all figures
```

Optional: regenerate model outputs (requires API keys):

```
python -m inference.run_questionnaire \  
    --models gpt-4o claude-sonnet-4-6 llama-3.3-70b  
python -m inference.run_qasper \  
    --models gpt-4o claude-sonnet-4-6 llama-3.3-70b
```

Human data availability. The human survey dataset is available in aggregated form only (option counts per demographic group); individual responses are not included to protect participant privacy. No personally identifiable information is present in any archived file.

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